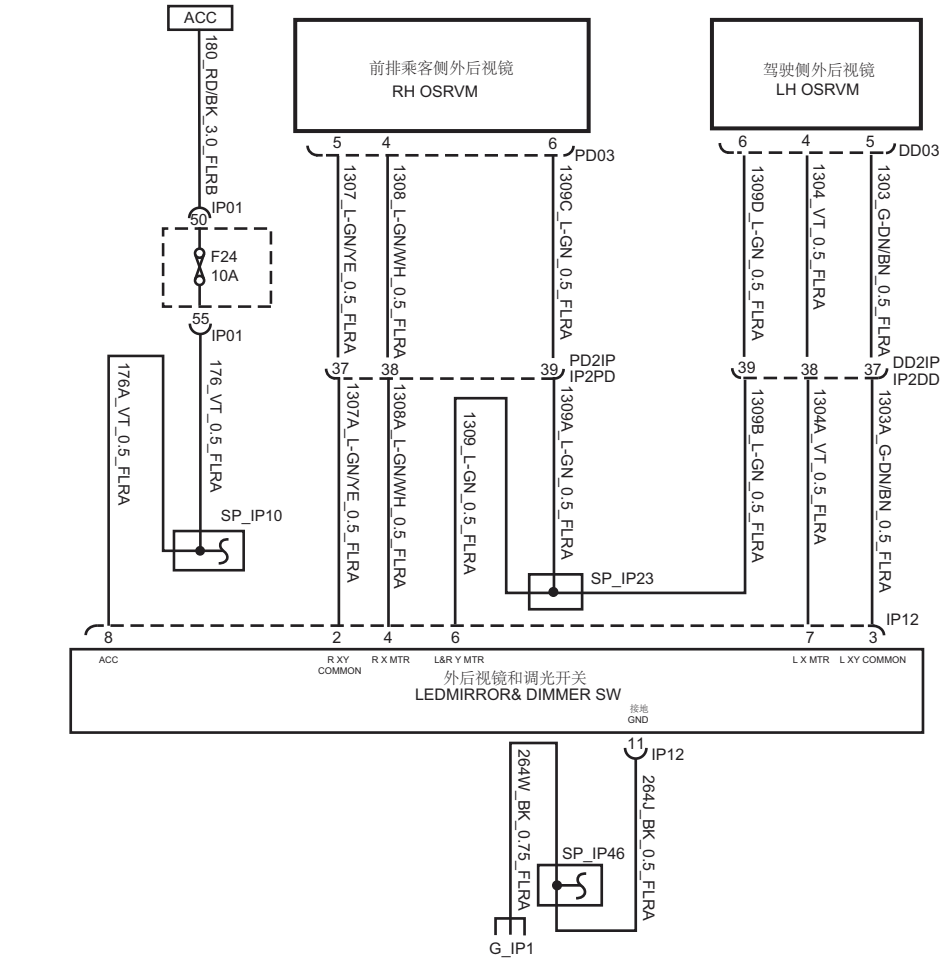
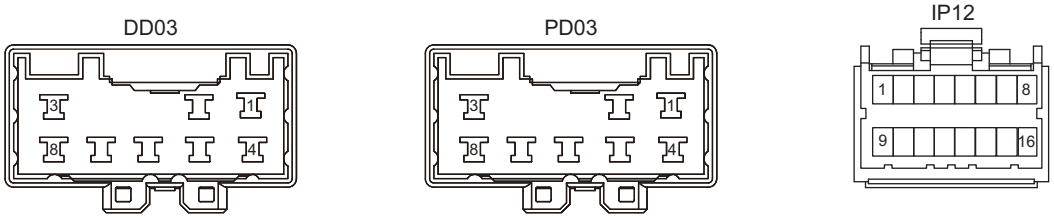
B1731-11、 B1731-12、 B1731-13、 B1732-11、 B1732-12、 B1732-13

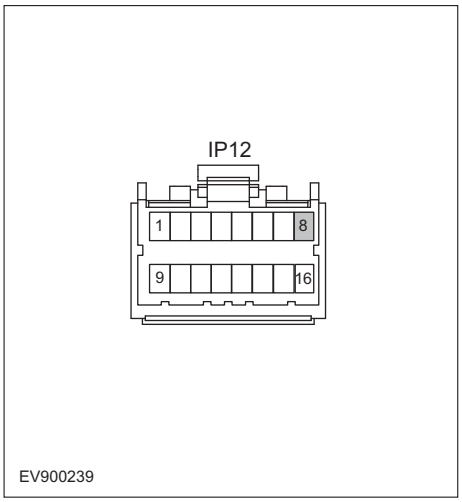
Fault diagnosis code
B1731-11：Rear mirror fold opening circuit short to ground
B1731-12：Rear mirror fold opening circuit short to battery
B1731-13：Rear mirror fold opening circuit open
B1731-11：Rear mirror fold closing circuit short to ground
B1731-12：Rear mirror fold closing circuit short to battery
B1731-13：Rear mirror fold closing circuit open
Detection tools
1 Multimeter and diagnostic apparatus
Possible causes
• Circuit,terminal and adapting piece fault • BCM Module fault • Rear view mirror fault • Insurance fuse fault
Conditions for fault code operation
Turn the ignition switch to ON position.
Conditions for fault code setting
B1731-11
Rear mirror fold opening circuit short to ground
B1731-12
Rear mirror fold opening circuit short to battery
B1731-13
Rear mirror fold opening circuit short to battery
B1732-11
Rear mirror fold opening circuit short to ground
B1732-12
Rear mirror fold opening circuit short to battery
B1732-13
Rear mirror fold closing circuit open
To set the effect of a fault code condition
1 Rear mirror fold opening circuit fault 2 Rear mirror fold closing circuit fault
Description of circuit diagram

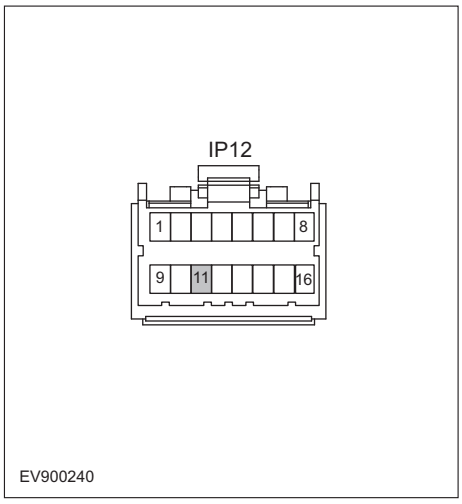
Circuit diagram

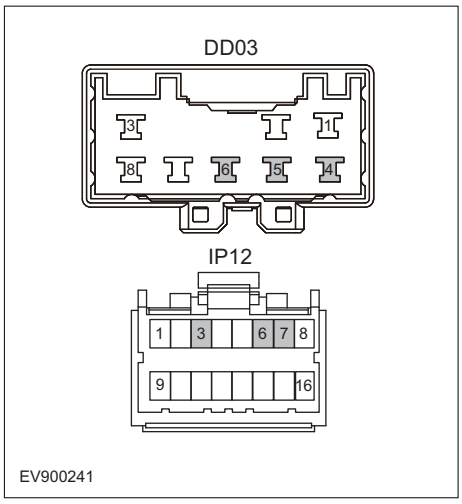


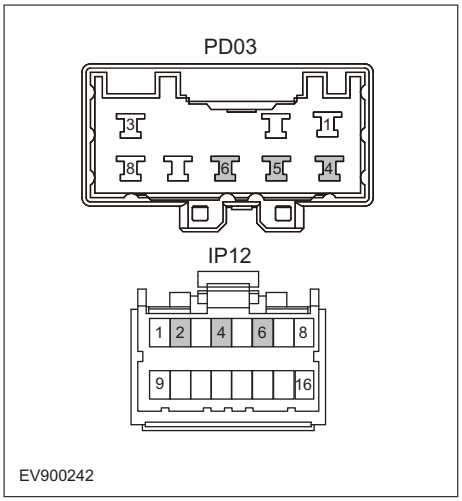


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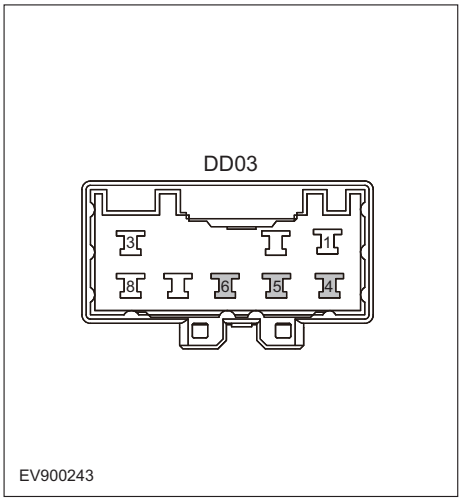
Troubleshooting steps		
Step 1	General inspection.	
		1 Check the fuse F24 for damage. 2 Check battery capacity. 1 Check the harness plug of rear view mirror and dimmer switch, BCM for damage, poor contact, aging and looseness. 2 Check whether the result is normal or not?
		NO → Repair or replace the faulted parts.
YES ↓		
Step 2	Check fault codes.	
		1 Connect the fault diagnosis instrument to the diagnosis interface. 2 Turn the ignition switch to ON position. 3 Diagnose the BCM with diagnostic apparatus. 4 If there is any fault code except for B1731-11、B1731-12、B1731-13、 B1732-11、 B1732-12、 B1732-13?
		YES → Refer to:DTC Summary list(BCM).
NO ↓		
Step 3	Check the supply power of rear view mirror and dimmer switch when it is started or operating.	
		1 Turn the ignition switch to OFF. 2 Disconnect the harness plug IP12 of rear view mirror and dimmer switch. 3 Turn the ignition switch to ON. 4 Measure the voltage between the harness terminal of rear view mirror and dimmer switch and ground. Measuring circuit: voltage between terminal 8 on plug IP12 and ground. **Standard value:10~14V** 5 Check whether the result is normal or not?
		NO → Repair or replace the faulted parts.
YES ↓		

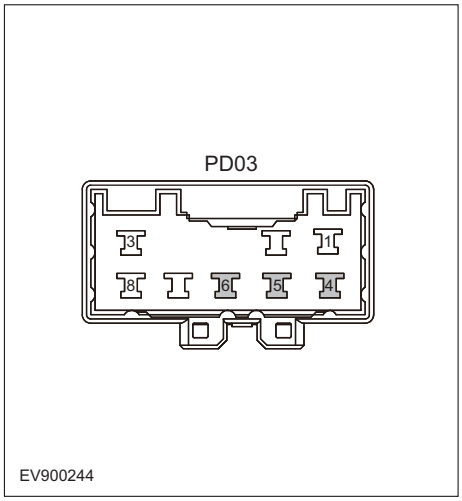
Step 4	Test whether the grounding circuit of rear view mirror and dimmer switch is open.
	1 Turn the ignition switch to OFF. 2 Measure the resistance between the harness terminal of rear view mirror and dimmer switch and ground. Measuring circuit: resistance between terminal 11 on plug IP12 and ground. **Standard value: < 1Ω** 3 Test whether the resistance is less than 1Ω?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 5	Check the circuit of the outside rear view mirror(open circuit).
	1 Turn the ignition switch to OFF. 2 Disconnect the harness plug DD03 of rear view mirror. 3 Measure the resistance between the harness terminal of outside rear view mirror from driver side and outside rear view mirror and dimmer switch and ground. Measuring circuit: resistance between terminal 4 on plug DD03 and terminal 7 on plug IP12. Measuring circuit: resistance between terminal 5 on plug DD03 and terminal 3 on plug IP12. Measuring circuit: resistance between terminal 6 on plug DD03 and terminal 6 on plug IP12. **Standard value: ≈ 0Ω** 4 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 6	Check the circuit of the front passenger outside rear view mirror(open circuit).
	1 Turn the ignition switch to OFF. 2 Disconnect the harness plug PD03 of side outside rear view of front row passenger. 3 Measure the resistance between the harness terminal of front passenger outside rear view mirror side and outside rear view mirror and dimmer switch and ground. Measuring circuit: resistance between terminal 4 on plug PD03 and terminal 4 on plug IP12. Measuring circuit: resistance between terminal 5 on plug PD03 and terminal 2 on plug IP12. Measuring circuit: resistance between terminal 6 on plug PD03 and terminal 6 on plug IP12. **Standard value: ≈ 0Ω** 4 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.

YES ↓

Step 7	Check the circuit of the driver outside rear view mirror(short circuit).
	1 Turn the ignition switch to OFF. 2 Disconnect the harness plug IP12 of outside rear view mirror and dimmer switch. 3 Measure the resistance between the harness terminal of driver outside rear view mirror and dimmer switch and ground. Measuring circuit: resistance between terminal 4 on plug DD03 and ground. Measuring circuit: resistance between terminal 5 on plug DD03 and ground. Measuring circuit: resistance between terminal 6 on plug DD03 and ground. **Standard value: ∞** 4 Measure the voltage between the harness terminal of driver outside rear view mirror and ground. Measuring circuit: voltage between terminal 4 on plug DD03 and ground. Measuring circuit: voltage between terminal 5 on plug DD03 and ground. Measuring circuit: voltage between terminal 6 on plug DD03 and ground. **Standard value: ≈ 0V** 5 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.

YES ↓	
Step 8	Check the circuit of the front passenger outside rear view mirror(short circuit).
	1 Turn the ignition switch to OFF. 2 Disconnect the harness plug IP12 of outside rear view mirror and dimmer switch. 3 Measure the resistance between the harness terminal of front passenger outside rear view mirror and dimmer switch and ground. Measuring circuit: resistance between terminal 4 on plug PD03 and ground. Measuring circuit: resistance between terminal 5 on plug PD03 and ground. Measuring circuit: resistance between terminal 6 on plug PD03 and ground. **Standard value: ∞** 4 Measure the voltage between the harness terminal of driver outside rear view mirror and ground. Measuring circuit: voltage between terminal 4 on plug PD03 and ground. Measuring circuit: voltage between terminal 5 on plug PD03 and ground. Measuring circuit: voltage between terminal 6 on plug PD03 and ground. **Standard value: ≈ 0V** 5 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 9	Replace the outside rear view mirror in good condition.
	1 Replace the outside rear view mirror in good condition. 2 Reconnect all disconnected wiring harness terminals. 3 Connect the fault diagnosis instrument to the diagnosis interface. 4 Turn the ignition switch to ON position. 5 Use the diagnostic apparatus to read BCM and DTC. 6 Clear the BCM and DTC. 7 Turn the ignition switch to OFF. 8 Waiting at least 15 seconds to allow the module to shut down properly. 9 Turn the ignition switch to ON again. 10 Use the diagnostic apparatus to read BCM and DTC again. 11 Test whether the faulted code is existed.
	NO → Replace the outside rear view mirror.
YES ↓	
Step 5	Test whether BCM is norm.
	1 Turn the ignition switch to OFF. 2 Disconnect and check the harness plug EB09_C、EB09_D、IP09_A、IP09_B of BCM. 3 Check and repair. • Corrosion(install new adapting piece or terminal-cleaning module and pin). • Pin damaged or bent-install new terminal/pin. • Fly out of pins-install new pins as needed. 4 Reconnect the BCM plug and all other previously disconnected plug. Make sure the pin is properly positioned and fixed. 5 Run the system and determine if the problem persists?
	YES → Change the BCM Module
NO ↓	
At this point the system is functioning normally. The problem may be caused by module connections. Address the root cause of any connector or pin problems,	
Next Step ↓	
Diagnosis end.	